Question Bank for CIE 1 22EC7BSBEC

1. The Frequency Modulated (FM) signal of commercial radios which is audible in Bengaluru could be classified as _____

- Ionizing Radiation
- Non-ionizing Radiation
- Nuclear Radiation
- Cosmic Radiation

2. Which among the following could be ionizing radiation

- WiFi signal operating in the 2.4GHz band
- 4G-LTE signals operating in the sub-6GHz bands
- GPS signals incoming from the geostationary satellites
- X-rays

3. Which among the following devices could be used to measure exposure to ionizing radiation ?

- Dosimeter
- Spectrum Analyzer
- Multimeter
- Optical Splicer

4. Which among these particles could contribute to ionizing radiation ?

- Neutrinos
- Muons
- Mesons
- Alpha particles

5. What is the unit of ionizing radiation measurement ?

- Sieverts
- Ohms
- Siemens
- Chernobyls

6. How much dosage of ionizing radiation is considered lethal ?

- 4-5 Sieverts
- 20-30 Sieverts
- 0.4-0.5 Sieverts
- 0.6-0.8 Sieverts

7. Arrange the penetrating power of the particles in ascending order

- Alpha (highest) Beta (moderate) Gamma (least)
- Beta (highest) Alpha (moderate) Gamma (least)
- Alpha (least) Beta (moderate) Gamma (highest)
- None of the above

8. A nuclear power plant in Harare, Zimbabwe has faced an explosion due to multiple issues. What would be an ideal strategy to contain radiation ?

- Bury the power plant with millions of tonnes of wood
- Burn the remaining power plant with Petrol or any other suitable inflammable agent
- Annihilate the power plant with 5 tonnes of TNT (Trinitrotoluene)
- Contain the power plant with a massive wall constructed with lead

9. Which among the following has the highest penetrating power ?

- Alpha
- Beta
- Gamma
- None of the above

10. A maniacal scientist has built a prototype for super high data rate with wireless technology. The wireless link offers a data rate of 1000 GBps. The only catch is that the carrier frequency of this technology is Gamma rays. Would you support this technology for commercialization?

- Yes, not a problem, millions of people will have flawless service
- Yes, but initial deployment cost would be high
- No, as gamma rays might be lethal for prolonged exposure
- I would stick with Non-Alignment movement as an irresponsible engineer

11. Half-life of industry grade Uranium used for commercial power generation is in the range of

- Billions of years
- Billions of light years
- Millions of years
- Millions of light years

12. Which among the following best describes Flouroscopy ?

- Uses x-rays and a fluorescent screen
- Uses Flourine gas to detect tissues
- Uses Radon gas along with Fluorescent screen for pictures
- None of the above

13. The risk of osteoporosis and bone fractures could be best diagnosed using which among these methods

- MRI
- CT scan
- Dual Energy X-ray Absorptiometry
- Trigonometry

14. Which among the following could be used as tracers for a Positron Emission tomography (PET) scan ?
- Uranium 235
- Helium 420
- Hydrogen 840
- Carbon 11C

15. Major exposure of natural source of radioactive material is from
- Injection
- Inflamation
- Exhalation
- Inhalation

Descriptive Questions

1. Explain ionizing and non-ionizing radiation (8 Marks)

2. Characterize alpha, beta and gamma radiations with various parameters (8 marks)

3. Discuss the biological damage effects as a consequence of radiation (7 Marks)